

Vulnerability Assessment Report

12th September 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database stored on the system holds critical information needed for key business operations. This data includes customer information which is used by employees to track customer activity and deliver a personalized shopping experience. The server is required to be accessible by remote employees to ensure continuity of marketing operations.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Perform reconnaissance and surveillance of organization	3	2	6
Hacker	Obtain sensitive information via exfiltration	3	3	9
System Administrator	Alter/Delete critical information	1	3	3

Approach

The risks were selected and evaluated based on the database's public exposure. Risks most likely to stem from this vulnerability were then assessed and quantified to determine their impact on company's normal business operations. The final risk score helps prioritize high impact threats from others.

Remediation Strategy

While current security measures – such as using latest OS version, SSL/TLS encrypted communications, and stable network configuration – are effective in maintaining overall system security, they do not mitigate risks posed by public exposure of database.

To further improve the security of system, it is necessary to enforce strict access control policies, including multi-factor authentication (MFA), to allow only authorized access. IP allow-listing is also another effective strategy to maintain confidentiality of database.